SOLUTION PITCH / CYBERSECURITY

ExposureGraph

Make an organization’s external attack surface easier to see, understand, and prioritize.

GNANI AI GREAT INDIAN INTERNSHIP CHALLENGE 2026
The idea: turn bounded, evidence-based external checks into a connected asset graph, prioritized findings, and a conversational assistant—with real Gnani speech APIs for voice input and spoken responses.

PROBLEM & USERS

Developers and small startup teams can lack a single, understandable view of internet-facing domains, subdomains, and web services. Scattered DNS, TLS, and HTTP observations are difficult to connect to affected assets and prioritize without dedicated security expertise.

PROPOSED SOLUTION

ExposureGraph is a web application for authorized external attack-surface assessment. A user supplies a domain and explicitly confirms authorization before a scan starts. The application discovers and checks public-facing web assets, records evidence and findings, calculates a deterministic exposure score, and presents connected assets and findings as an interactive graph.

WORKFLOW

Authorize target → discover domain and common subdomains via DNS → validate public IPs → inspect TLS and bounded web services → save evidence, assets, and findings → graph and risk view → ask follow-up questions by text or voice.

USEFUL & DISTINCTIVE

• Connects findings to the discovered asset graph, not just a list of alerts.
• Uses a transparent, deterministic severity/confidence risk model; the assistant answers scan questions from saved evidence.
• Combines text interaction with voice input and audio responses through live backend integrations with Gnani Prisma STT and Timbre TTS.
• Includes authorization gates and bounded checks: non-public IPs are rejected and service probes are limited to ports 80, 443, 8080, and 8443.

EXPECTED IMPACT & FUTURE SCOPE

Expected: help lean teams build a clearer picture of their public web footprint and use recorded evidence to decide what to investigate first. No outcome or performance metrics are claimed.
Next: validate with authorized users, broaden assessment coverage carefully, and develop a production deployment path. These are future directions, not current capabilities.

FEATURES & TECHNICAL ARCHITECTURE

• Asset inventory and React Flow graph of domains, subdomains, IPs, services, relationships, and attached findings.
• DNS discovery; bounded HTTP/TLS checks; evidence-backed findings and severity/confidence-based risk scoring.
• Authenticated dashboard and assistant for scan initiation and scan follow-up.
• Gemini replies use scan tools and evidence; deterministic fallback is available.

FRONTEND

React 19, TypeScript, Vite, React Flow

BACKEND

Python, FastAPI, SQLAlchemy, Alembic; SQLite for local development and PostgreSQL support; dnspython and httpx for DNS and HTTP integrations

ASSISTANT

Google Gen AI SDK with Gemini 2.5 Flash for evidence-grounded conversational replies when configured

GNANI AI INTEGRATIONS IN THE CODEBASE

Prisma STT · speech to text
The authenticated voice-transcription API accepts audio uploads and calls Gnani’s Prisma REST endpoint (**api.vachana.ai/stt/v3**) using the X-API-Key-ID header and multipart audio. Configurable defaults: model **prisma-v3.0**, language code **indic-auto**. The transcript and detected language feed the assistant workflow.

Timbre TTS · text to speech
The authenticated synthesis API sends assistant text to Gnani’s Timbre REST endpoint (**api.vachana.ai/api/v1/tts/inference**) with the X-API-Key-ID header. Configurable defaults: model **timbre-v2.5**, voice **Nalini**, language **en-IN**. Audio is returned as WAV; a streaming route synthesizes text chunks concurrently.

Both integrations require **GNANI_API_KEY** in the runtime environment; credentials are not embedded in the application source.

SCOPE & RESPONSIBLE USE

The scanner is best-effort and is not a penetration test. It checks a bounded set of common subdomains and public web services; a timeout or unavailable host is not proof of a vulnerability. Scans are intended only for domains the user owns or is authorized to assess.